Mid-Semester Examination Solutions

Markov Chains and Queuing Systems (EE 621), Spring'26

QUESTION 1

The stationary distribution is the solution of the system of equations $\pi = \pi P$ and $\sum_{i=0}^{2} \pi_i = 1$, where π_i is the stationary probability of state i. Solving this system of equations yields $\pi_0 = \frac{21}{62}$, $\pi_1 = \frac{23}{62}$, and $\pi_2 = \frac{18}{62}$.

QUESTION 2

Suppose a renewal starts each time the worker starts working on a new job. Let $X_1, X_2, X_3, \ldots$ be the interarrival times. Let Y_i be the completion time of the ith job. Also, let Z_i be the time until the next shock occurs after work on job i starts. Then Z_i is exponential(λ).

Let the reward

$$R_i = \begin{cases} 1, & \text{if the } i\text{th job is complete,} \\ 0, & \text{else.} \end{cases}$$

Then the reward, $R(t)$, obtained up to time t is the number of jobs completed up to time t. By the renewal reward theorem, the long run rate at which jobs are completed is

$$\lim_{t\to\infty} \frac{R(t)}{t} = \frac{E(R_i)}{E(X_i)}.$$

But,

$$E(R_i) = P(i\text{th job is completed}) = \int_0^\infty e^{-\lambda y} dF(y).$$

Also, $X_i = \min(Y_i, Z_i)$. Hence, $X_i > x$ iff $Y_i > x$ and $Z_i > x$. So,

$$P(X_i > x) = P(Y_i > x)P(Z_i > x) = F^c(x)e^{-\lambda x}.$$

Now,

$$E(X_i) = \int_0^\infty P(X_i > x)dx = \int_0^\infty F^c(x)e^{-\lambda x}dx.$$

Thus, in the long run, jobs are completed at the rate

$$\frac{E(R_i)}{E(X_i)} = \frac{\int_0^\infty e^{-\lambda y} dF(y)}{\int_0^\infty F^c(x)e^{-\lambda x}dx}.$$

QUESTION 3

(a) Given that $N(t) = n$, each arrival is uniformly distributed in $(0, t)$; so,

$$E(X|N(t) = n) = n \int_0^t (t - s) \frac{ds}{t} = \frac{nt}{2}.$$

(b) Let $U_1, U_2, \ldots$ be independent uniform $(0, t)$ random variables. Then

$$\text{var}(X|N(t) = n) = \text{var}\left[\sum_{i=1}^{n} (t - U_i)\right] = n\text{var}(U_i) = \frac{nt^2}{12}.$$

(c) It is easy to check that the following result, which is known as the conditional variance formula, holds for all random variables X and Y:

$$\text{var}(X) = E\left[\text{var}(X|Y)\right] + \text{var}(E(X|Y)).$$

By parts (a) and (b) and the conditional variance formula, we get:

$$\text{var}(X) = \text{var}\left(\frac{N(t)t}{2}\right) + E\left[\frac{N(t)t^2}{12}\right] = \frac{\lambda t t^2}{4} + \frac{\lambda t t^2}{12} = \frac{\lambda t^3}{3}.$$

QUESTION 4

We use a procedure similar to that in slides 20 and 21 of "Discrete-Time Markov Chains (DTMCs): Part 1", with the main differences being as follows.

The n-step probability of return to 0 is

$$P_{0,0}^n = \begin{cases} \begin{pmatrix} n \\ n/3 \end{pmatrix} p^{n/3}(1-p)^{2n/3}, & \text{if } n \in \{3, 6, 9, \ldots\}, \\ 00, & \text{else.} \end{cases}$$

The DTMC is transient iff the following quantity is finite:

$$S = \sum_{m=1}^{\infty} \begin{pmatrix} 3m \\ m \end{pmatrix} p^m (1-p)^{2m}.$$

Using Stirling's approximation and simplifying, we get

$$S \approx \sum_{m=1}^{\infty} \sqrt{\frac{3}{4\pi m}} \left[\frac{27}{4} p(1-p)^2\right]^m. \tag{1}$$

Letting $f(p) = \frac{27}{4} p(1-p)^2$, it is easy to show that $f(p) = 1$ for $p = 1/3$ and $f(p) < 1$ for $p \in (0, 1)$, $p \neq 1/3$. So, when $p = 1/3$ (respectively, $p \in (0, 1)$, $p \neq 1/3$), the series in (1) diverges (respectively, converges) and hence the DTMC is recurrent (respectively, transient).

QUESTION 5

(a) Note that the number of children in family i who are jth born and have l younger siblings equals 1 if $\nu_i = j + l$ and 0 else. So, the required fraction is

$$\lim_{n\to\infty} \frac{\sum_{i=1}^{n} \mathbf{1}_{\{\nu_i = j+l\}}}{\sum_{i=1}^{n} \nu_i},$$

which is equal to $\frac{\alpha_{j+l}}{E(\nu)}$ w.p. 1 by the SLLN.

(b) (i) The fraction of children who are jth born is

$$\frac{\sum_{l=0}^{\infty} \alpha_{j+l}}{E(\nu)}.$$

(ii) The fraction of children who have l younger siblings is

$$\frac{\sum_{j=1}^{\infty} \alpha_{j+l}}{E(\nu)}.$$

(c)

$$P(K = k) = \frac{\sum_{j=1}^{k} \alpha_k}{E(\nu)} = \frac{k\alpha_k}{E(\nu)}.$$

QUESTION 6

If we say that the system is on when there is at least one customer in the system, and off when the system is empty, then we have an alternating renewal process. We have $\int_0^\infty G^c(t)dt = E(S)$, where $E(S)$ is the mean of the service distribution G. From the results of the example on slide 24 of "The Poisson Process", we get that the number of customers in the system at time t is Poisson distributed with mean $\lambda \int_0^t G^c(y)dy$. So, the probability that the system is off at time t is $e^{-\lambda \int_0^t G^c(y)dy}$. Hence,

$$\lim_{t\to\infty} P\{\text{system is off at time } t\} = e^{-\lambda E(S)}.$$

So,

$$e^{-\lambda E(S)} = \frac{E(\text{off time in cycle})}{E(\text{duration of a cycle})}.$$

But when the system goes off, it remains off only up to the time of the next arrival, giving that

$$E(\text{off time in cycle}) = \frac{1}{\lambda}.$$

Also,

$$E(\text{on time in cycle}) = E(B),$$

where $E(B)$ is the expected length of a busy period. Hence,

$$e^{-\lambda E(S)} = \frac{1/\lambda}{1/\lambda + E(B)},$$

which yields

$$E(B) = \frac{1}{\lambda}\left(e^{\lambda E(S)} - 1\right).$$

QUESTION 7

$X(t)$ is a regenerative process, which regenerates itself whenever a truck arrives. Hence,

$$\lim_{t\to\infty} P(X(t)=i) = \frac{E\left(\text{Amount of time for which } X(t)=i \text{ during one cycle}\right)}{E\left(\text{Length of a cycle}\right)} \tag{2}$$

Now,

$$E\left(\text{Length of a cycle}\right) = \int_0^\infty F^c(z)dz. \tag{3}$$

Let Z be the interarrival time of trucks. Then

$$\begin{aligned}
&E\left(\text{Amount of time for which } X(t)=i \text{ during one cycle}\right)\\
&= E\left(\int_0^Z \mathbf{1}_{\{X(t)=i\}}dt\right)\\
&= E\left(\int_0^\infty \left(\int_0^z \mathbf{1}_{\{X(t)=i\}}dt\right) dF(z)\right)\\
&= \int_0^\infty \left(\int_0^z P(\{X(t)=i\})dt\right) dF(z)\\
&= \int_0^\infty \int_0^z \frac{e^{-\lambda t}(\lambda t)^i}{i!}dtdF(z)
\end{aligned} \tag{4}$$

The required expression can be obtained by substituting (3) and (4) into the RHS of (2).

QUESTION 8

(a) This can be shown similarly to our proof of the corollary of Wald's equation on slide 21 of "Renewal Theory".

(b) Suppose $0 < p < 1$ and

$$P(X_1 = a) = \begin{cases} p, & \text{if } a = 1,\\ 1-p, & \text{if } a = 2. \end{cases}$$

Then $\mu = 2 - p \neq 1$. Also

$$E(S_{N(1)}) = (1-p)E(S_0|N(1)=0) + pE(S_1|N(1)=1) = p,$$

whereas $m(1) = p$. Hence, $E(S_{N(1)}) \neq \mu m(1)$.

QUESTION 9

See Fig. 1.

To determine $P(N \leq 8)$, define a Markov chain with states 0, 1, 2, 3 where for $i < 3$ state i means that we currently are on a run of i consecutive heads, and where state 3 means that a run of three consecutive heads has already occurred. Thus, the transition probability matrix is

$$\mathbf{P} = \begin{pmatrix} 1/2 & 1/2 & 0 & 0 \\ 1/2 & 0 & 1/2 & 0 \\ 1/2 & 0 & 0 & 1/2 \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

where, for instance, the values for row 2 are obtained by noting that if we currently are on a run of size 1 then the next state will be 0 if the next flip is a tail, or 2 if it is a head. Hence, $P_{1,0} = P_{1,2} = 1/2$. Because there would be a run of three consecutive heads within the first eight flips if and only if $X_8 = 3$, the desired probability is $P^8_{0,3}$. Squaring $\mathbf{P}$ to obtain $\mathbf{P}^2$, then squaring the result to obtain $\mathbf{P}^4$, and then squaring that matrix gives the result

$$\mathbf{P}^8 = \begin{pmatrix} 81/256 & 44/256 & 24/256 & 107/256 \\ 68/256 & 37/256 & 20/256 & 131/256 \\ 44/256 & 24/256 & 13/256 & 175/256 \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

Hence, the probability that there will be a run of three consecutive heads within the first eight flips is $107/256 \approx 0.4180$.

(b) Noting that $N = 8$ if the pattern hasn't yet occurred in the first 7 transitions, the state after 7 transitions is 2, and the next flip lands heads, shows that

$$P(N = 8) = \frac{1}{2} P^7_{0,2}.$$

Fig. 1. The solution for Question 9.